

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 95-029

WASTE DISCHARGE REQUIREMENTS
FOR
TERRA BELLA SEWER MAINTENANCE DISTRICT
WASTEWATER TREATMENT FACILITY
TULARE COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. The Terra Bella Sewer Maintenance District (hereafter Discharger) submitted a Report of Waste Discharge, dated 26 June 1993, to revise requirements for its domestic wastewater treatment facility serving the community of Terra Bella.
2. Waste Discharge Requirements Order No. 85-172, adopted by the Board on 28 June 1985, prescribes requirements for a discharge of up to 0.1 mgd of domestic waste from a wastewater treatment facility consisting of a single aeration lagoon to six evaporation/percolation ponds and adjacent pastures and orchards. The treatment facility has been modified and currently consists of a bar screen, grinder, Parshall flume, an aeration lagoon, two stabilization ponds, a chlorine contact chamber, and seven disposal ponds. The current thirty-day average daily wastewater flow is 0.212 million gallons. The chlorine contact chamber is currently not in use.
3. Order No. 85-172 no longer reflects the wastewater treatment facility and it is neither adequate nor consistent with current plans and policies of the Board.
4. The facility is in Section 34, T22S, R27E, MDB&M, as shown on Attachment A, a part of this Order.
5. The Terra Bella wastewater treatment facility has been operating at or above its maximum capacity for several years and has experienced pond embankment failures which resulted in an illegal release of partially treated wastewater off-site from 11 March 1992 to 1 April 1992. On 26 June 1992, the Board adopted Cease and Desist Order No. 92-113 requiring the Discharger to cease discharging waste contrary to its requirements. Order No. 92-113 directed the Discharger to improve its facility to provide additional treatment and flow capacity by 30 January 1994. Time schedules stipulated in Order No. 92-113 have expired and the Discharger has completed sufficient facility modifications to increase treatment and disposal capacity to handle current flows.

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6. The Discharger has submitted plans to modify the existing facility further for additional treatment and disposal capacity at this site. The proposed modifications and new construction will increase total capacity to 0.3 mgd. The expanded wastewater treatment facility will consist of a bar screen, grinder, Parshall flume, two aeration lagoons, four stabilization ponds, four evaporation/percolation ponds, and two storage ponds for reclamation. Wastewater reclamation will be initiated to reuse wastewater whenever possible. However, design calculations show there will be sufficient disposal capacity for a flow of 0.3 mgd with the disposal ponds alone. The Discharger is in the process of entering into an agreement with a local farmer to purchase additional land adjacent to the facility and is currently securing funds from the Farmers Home Administration's Loan/Grant Program to complete modifications to the facility. Construction for the entire modified facility is expected to be completed in October 1996.
7. The Discharger is in the process of entering into an agreement with Albert Bakalian to reclaim treated wastewater for surface irrigation of about 52 acres of pistachio, orange, and olive orchards adjacent to the facility. Albert Bakalian will submit a separate Report of Reclamation to the Board prior to reclaiming wastewater.
8. The California Department of Health Services has established statewide reclamation criteria in Title 22, California Code of Regulations (CCR), Division 4, Chapter 3 (hereafter Title 22) for the use of reclaimed water and developed guidelines for specific uses.
9. Soils beneath the site consist of non-marine alluvial deposits of mixed grain size with moderate rates of infiltration and permeability. Average annual precipitation and pan evaporation at this facility are 11 and 75 inches, respectively.
10. The site lies within the Tule Delta Hydrologic Subarea (No. 558.20) in the South Valley Floor Hydrologic Unit, as defined on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.
11. Surface water drainage is to Fountain Springs Gulch, a tributary to Deer Creek, about 1.2 miles downstream of the wastewater treatment facility. Deer Creek is a valley floor water.
12. The Board adopted a Water Quality Control Plan for the Tulare Lake Basin (hereafter Basin Plan) which contains water quality objectives for all waters of the Basin. These requirements implement the Basin Plan.
13. The designated beneficial uses of Valley Floor Waters are agricultural and industrial supply, water contact and non-contact water recreation, warm fresh water habitat, wildlife habitat, preservation of rare and endangered species, and ground water recharge.

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14. Community water supply is from local wells, Deer Creek, and the Friant-Kern Canal. Depth to ground water in the area was reported in spring 1983 at 155 feet below ground surface. Ground water quality is good (EC of about 500 $\mu\text{mhos/cm}$). However, sample results of on-site monitoring wells show shallow ground water quality beneath the site is fair (EC of 773 to 976 $\mu\text{mhos/cm}$). Sampling of influent wastewater at the facility revealed an average specific electrical conductivity between 424 and 470 $\mu\text{mhos/cm}$. The beneficial uses of underlying ground water are municipal and domestic, industrial, and agricultural supply.
15. On 16 November 1993 the Terra Bella Sewer Maintenance District certified a negative declaration for this project in accordance with the California Environmental Quality Act (Public Resources Code, Section 21000, et seq.), and State CEQA Guidelines.
16. The permitted discharge is consistent with the antidegradation provisions of State Water Resources Control Board Resolution 68-16. The project, as approved, will not have a significant effect on water quality.
17. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
18. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Waste Discharge Requirements Order No. 85-172 is rescinded and the Terra Bella Sewer Maintenance District, its agents, successors and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions

1. Discharge of waste to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated domestic wastewater is prohibited.
3. Discharge of waste classified as 'hazardous' or 'designated', as defined in Sections 2521(a) and 2522(a) of Title 23, CCR, Section 2510, et seq. (hereafter Chapter 15), is prohibited.

B. Discharge Specifications

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4. The monthly average dry weather discharge flow shall not exceed 0.22 million gallons per day. Upon acceptance of certification pursuant to Provision E.4, the monthly average dry weather discharge flow shall not exceed 0.3 mgd.
5. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the Discharger's property.
6. As a means of discerning compliance with Discharge Specification B.2, the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l.
7. The wastewater treatment facility shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
8. Effluent from the treatment facility shall not contain constituents in excess of the following limits:

<u>Constituent</u>	<u>Units</u>	<u>Monthly Average</u> ¹	<u>Daily Maximum</u> ¹
BOD ₅ ²	mg/l	40	80
Suspended Solids	mg/l	40	80

¹ Effluent limits for these constituents are established at the point of compliance (POC). The POC for this facility is after the stabilization pond and before discharge to the evaporation/percolation ponds or storage ponds.

² Five-day, 20° Celsius biochemical oxygen demand.

9. The maximum electrical conductivity (EC) of the discharge shall not exceed 900 µmhos/cm or the average EC of the source water plus 500 µmhos/cm, whichever is less.
10. The effluent in the stabilization and disposal ponds shall not have a pH less than 6.5 or greater than 9.0.
11. Public contact with wastewater shall be precluded through such means as fences, signs, or other acceptable alternatives.

12. Ponds shall be managed to prevent breeding of mosquitos. In particular:
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
13. Ponds shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than two feet (measured vertically).
14. On or about 1 October of each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specification B.10.
15. Use of reclaimed water shall be limited to surface irrigation of orchards, groves, and vineyards, and fodder, fiber, and seed crops of sites with valid water reclamation requirements.

C. Solids Disposal Specifications

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Chapter 15, and approved by the Executive Officer.
2. Any proposed change in solids use or disposal practice shall be reported to the Executive Officer **at least 90 days in advance** of the change.
3. Use and disposal of sewage sludge shall comply with existing federal and state laws and regulations, including permitting requirements and technical standards included in 40 CFR 503.

If the State Water Resources Control Board and the Regional Water Quality Control Boards implement regulations contained in 40 CFR 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger must comply with the standards and time schedules contained in 40 CFR 503 whether or not they have been incorporated into this Order.

D. Ground Water Limitations

The discharge, in combination with other sources, shall not cause underlying ground water to:

16. Contain waste constituents in concentrations statistically greater than receiving water limits, where specified below, or background water quality where not specified. (For purposes of comparison, background water quality shall be determined when background monitoring provides sufficient data. Quality determined in this manner establishes "water quality protection standards.")
17. Exceed an annual average incremental increase in specific electrical conductivity greater than 6 $\mu\text{mhos/cm}$, based on the most recent five-year period, or a maximum of 1000 $\mu\text{mhos/cm}$, whichever is less.
18. Contain chemicals, heavy metals, or trace elements in concentrations that adversely affect beneficial uses identified in Finding No. 14 or exceed maximum contaminant levels specified in 22 CCR, Division 4, Chapter 15.
19. Exceed a most probable number of total coliform organisms of 2.2/100 ml over any seven-day period.
20. Exceed concentrations of radionuclides specified in 22 CCR, Division 4, Chapter 15.
21. Contain taste or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses.

E. Provisions

22. The Discharger shall comply with attached Monitoring and Reporting Program No. 95-029, which is part of this Order, and any revisions thereto, as ordered by the Executive Officer.
23. The Discharger shall comply with all the items of the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated 1 March 1991, which are part of this Order. This attachment and its individual paragraphs are referred to as "Standard Provision(s)."

24. In the event of any change in control or ownership of land or the wastewater treatment facility described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.

To assume operation under this Order, the succeeding owner or operator must apply in writing to the Executive Officer requesting transfer of the Order. The request must contain the requesting entity's full legal name, the State of incorporation if a corporation, the name and address and telephone number of the persons responsible for contact with the Board, and a statement. The statement shall comply with the signatory paragraph of Standard Provision B.3 and state that the new owner or operator assumes full responsibility for compliance with this Order. Failure to submit the request shall be considered a discharge without requirements, a violation of the California Water Code. Transfer shall be approved or disapproved in writing by the Executive Officer.

25. Once long-term project improvements are complete and the facilities operational, the Discharger shall submit a written report with certification from a registered engineer that the wastewater treatment facility is complete and able to adequately treat and dispose of 0.3 mgd of wastewater. When certification of completion is received and accepted by the Executive Officer in writing, the flow limit will be increased to 0.3 mgd.
26. The Discharger shall use the best practicable cost-effective control technique currently available to comply with salinity limits specified in this order.
27. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
28. A copy of this Order shall be kept at the discharge facility for reference by the personnel responsible for operating and maintaining the treatment facility. Key operating personnel shall be familiar with its contents.
29. If reclaimed water is used for construction purposes, it shall comply with the most current edition of "Guidelines for Use of Reclaimed Water for Construction Purposes". Other uses of reclaimed water not specifically authorized herein shall be subject to the approval of the Executive Officer and shall comply with 22 CCR, Division 4.

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30. The Board will review this Order periodically and will revise requirements when necessary.

I, WILLIAM H. CROOKS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 24 February 1995.

WILLIAM H. CROOKS, Executive Officer

JGW:cjs: 2/24/95

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 95-029

FOR
TERRA BELLA SEWER MAINTENANCE DISTRICT
WASTEWATER TREATMENT FACILITY
TULARE COUNTY

Specific sample station locations shall be established with concurrence of the Board's staff, and a description of the stations shall be submitted to the Board and attached to this program.

INFLUENT MONITORING

Influent samples shall be collected at the inlet of the headworks and approximately the same time as effluent samples. Influent monitoring shall include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Flow	mgd	Metered	Continuously
BOD ₅ ¹	mg/l	Grab	Weekly
Total Suspended Solids	mg/l	Grab	Weekly

¹ Five-day, 20° Celsius biochemical oxygen demand.

EFFLUENT MONITORING

Effluent samples shall be collected just prior to discharge to the disposal ponds or reclamation storage ponds. Effluent samples should be representative of the volume and nature of the discharge. Samples collected from the outlet structure of the stabilization ponds will be considered adequately composited. Time of collection of grab sample shall be recorded. Effluent monitoring shall include the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency¹</u>
BOD ₅ ²	mg/l	Grab	Weekly
pH	pH Units	Grab	Weekly

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<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u> ¹
Total Suspended Solids	mg/l	Grab	Weekly
Nitrates as N	mg/l	Grab	Quarterly
Total Dissolved Solids	mg/l	Grab	Quarterly
Electrical Conductivity	µmhos/cm	Grab	Quarterly

¹ If results of monitoring a pollutant appear to violate effluent limitations, but monitoring frequency is not sufficient to validate violation (e.g., the monthly average for BOD), or indicate a violation and potential upset of the treatment process (e.g., less than minimum D.O.), the frequency of sampling shall be increased to confirm the magnitude and duration of violation, if any, and aid in identification and resolution of the problem.

² Five-day, 20° Celsius biochemical oxygen demand.

POND MONITORING

The freeboard shall be monitored on all ponds to the nearest tenth of a foot. Monitoring of ponds shall include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Dissolved Oxygen ¹	mg/l	Grab	Weekly
Freeboard	feet	Observation	Weekly
pH	pH Units	Grab	Weekly

¹ Samples will be collected from the opposite of the inlet to each pond and analyzed for dissolved oxygen. Samples will be collected between 0800 and 0900 hours.

A permanent marker shall be placed in each pond with calibration indicating the water level at design capacity and available operational freeboard.

In addition, the Discharger shall inspect the condition of the pond once per week and write visual observations in a bound log book. Notations shall include observations of whether weeds are developing in the water or along the bank, and their location; whether dead algae, vegetation, scum, or debris are accumulating on the pond surface and their location; whether burrowing animals or insects are present; and the color of the pond (e.g., dark sparkling green, dull green, yellow, grey, tan, brown, etc.). A copy of the entries made in the log during each month shall be submitted along with the monitoring report the following month. Where the O&M manual indicates remedial action is necessary, the Discharger shall briefly explain in the transmittal what action has been taken or is scheduled to be taken.

REPORTING

Monitoring results shall be submitted to the Board by the **20th day of the month** following sample collection.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner to illustrate clearly whether the discharge complies with waste discharge requirements, including calculation of all averages, etc.

If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the calculation and reporting of the values required in the discharge monitoring report form. Such increased frequency shall be indicated on the discharge monitoring report form.

By **31 January of each year**, the Discharger shall submit a written report to the Executive Officer containing the following:

- a. The names, titles, certificate grade (if any) and general responsibilities of persons operating and maintaining the wastewater treatment facility.
- b. The names and telephone numbers of persons to contact regarding the facility for emergency and routine situations.
- c. A certified statement of when the flow meter and other monitoring instruments and devices were last calibrated, including identification of who did the calibration (Standard Provision C.4).
- d. A statement certifying whether the current operation and maintenance manual, and contingency plan, reflect the treatment facility as currently constructed and operated, and the dates when these documents were last revised and last reviewed for adequacy.

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- e. The total quantity of sludge disposed of during the previous year and ultimate disposal site(s).

The Discharger may also be requested to submit an annual report to the Board with both tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss the compliance record. If violations have occurred, the report shall also discuss the corrective actions taken and planned to bring the discharge into full compliance with the waste discharge requirements.

All reports submitted in response to this Order shall comply with the signatory requirements of Standard Provision B.3.

The Discharger shall implement the above monitoring program on the first day of the month following the effective date of this Order.

Ordered by: _____
WILLIAM H. CROOKS, Executive Officer

24 February 1995
(Date)

JGW:cjs: 2/24/95

INFORMATION SHEET

TERRA BELLA SEWER MAINTENANCE DISTRICT WASTEWATER TREATMENT FACILITY TULARE COUNTY

The Terra Bella Sewer Maintenance District (Discharger) submitted a Report of Waste Discharge, dated 26 June 1993, to revise requirements for its domestic wastewater treatment facility in Section 34, T22S, R27E, MDB&M. The community of Terra Bella has a current population of about 1,800 people. Sewer hookups to the treatment facility is made up of about 94 percent residential and 6 percent commercial/industrial. The average monthly flow reported between July 1994 and October 1994 was 0.212 mgd, with an average monthly flow of 0.22 mgd during August 1994.

The facility is currently regulated by Waste Discharge Requirements Order No. 85-172, adopted by the Board on 28 June 1985, which prescribes requirements for a discharge of up to 0.1 mgd of domestic waste.

The wastewater facility has been operating at or above its maximum capacity for several years and has experienced pond embankment failures which resulted in an illegal release of partially treated wastewater off-site. On 26 June 1992, the Board adopted Cease and Desist Order No. 92-113 requiring the Discharger to cease discharging waste contrary to its requirements. Order No. 92-113 directed the Discharger to improve its facility to provide additional treatment and flow capacity by 30 January 1994. The Discharger has completed construction of improved headworks, increased the number and size of disposal ponds, and initiated interim reclamation on adjacent olive orchards to handle current flows.

The Discharger has submitted plans to further modify the existing facility and construct additional treatment and disposal capacity at this site. The proposed modifications and new construction will require the purchase of additional land and will increase total treatment capacity to 0.3 mgd. The Discharger has applied, and has been approved, for loans and grants from the Farmers Home Administration to modify the current facility. Construction for the entire modified facility is expected to be completed in October 1996.

The Discharger is in the process of entering into an agreement with a local farmer to purchase 20 acres of adjacent property for facility expansion and to supply reclamation water for agricultural use. The farmer has been advised to submit a Report of Water Reclamation to the Board prior to reclaiming wastewater from the Discharger.

Discharge Specification B.1 of these requirements limits the monthly average daily discharge flow of wastewater to 0.22 million gallons until the Board has received written certification that expansion and modifications of the treatment facilities is complete. The Discharger has requested a current limit of 0.25 mgd. The limit of 0.22 mgd, though only slightly greater than current flows, is in the opinion of staff, all the current facility is capable of adequately handling based on a site inspection of 17 January 1995 and review of technical information provided by

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the Discharger. Upon acceptance of certification by the Executive Officer, the monthly average discharge flow will be increased to 0.3 mgd.

Soils beneath the site consist of non-marine alluvial deposits of mixed grain size with moderate rates of infiltration and permeability.

Community water supply is from local wells, Deer Creek, and the Friant-Kern Canal. Depth to ground water in the area was reported in spring 1983 at 155 feet below ground surface. Ground water quality is good (EC of about 500 $\mu\text{mhos/cm}$). However, sampling of on-site monitoring wells show shallow ground water quality beneath the site is fair (EC about 773 to 976 $\mu\text{mhos/cm}$). The beneficial uses of underlying ground water are municipal and domestic, industrial, and agricultural supply.

Surface water drainage is to Fountain Springs Gulch, a tributary to Deer Creek, about 1.2 miles downstream of the wastewater treatment facility. Deer Creek is a valley floor water. The designated beneficial uses of Valley Floor waters are agricultural and industrial supply, water contact and non-contact water recreation, warm fresh water habitat, wildlife habitat, preservation of rare and endangered species, and ground water recharge.

The Terra Bella Sewer Maintenance District has certified a negative declaration for this project in accordance with the California Environmental Quality Act (Public Resources Code, Section 21000, et seq.), and State Guidelines. The project, as approved, will not have a significant effect on water quality.